

Creative Programming: Resources

Lecture 10

Narration

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(Re)visiting slides?

You can jump to any slide with the menu to the (bottom) left.

Did you miss a slide or want to revisit?

Open the narration tab while studying to get an explanation of difficult slides.



Today's lecture

- Last lecture!!!
- Today is a shorter, resource-focused lecture
- No programming - just some inspiration and perhaps project work time
- We explore **five p5.js libraries** to serve as inspiration for your projects
- Focus on what excites you

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What is a library?

- A library is extra code that gives p5.js **new abilities**
- We've used one before: p5js sound
- You include it with a `<script>` tag in your HTML
- Then you can use its new functions in your sketch

```
1 <html>
2   <head>
3     <script src="https://cdn.jsdelivr.net/npm/p5/lib/p5.min.js"></script>
4     <!-- Add your library here -->
5     <script src="https://the-library-url.js"></script>
6   </head>
7   <body>
8     <script src="sketch.js"></script>
9   </body>
10 </html>
```

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Games and Physics

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Why games and physics?

- Games are engaging but also hard to make
- They combine visuals, interaction, and logic
- Even simple games can be incredibly fun to build and play

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What is p5play?

- A game engine built on p5.js
- Powered by the **Box2D** physics engine (same as Angry Birds!)
- Designed by CS teachers for beginners
- Handles gravity, collisions, sprites, animation, input
- p5play.org

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Including p5play

Download both files into your project folder (preferred):

- [planck.min.js](#)
- [p5play.js](#)

```
1 <script src="planck.min.js"></script>  
2 <script src="p5play.js"></script>
```

Or load from the web:

```
1 <script src="https://p5play.org/v3/planck.min.js"></script>  
2 <script src="https://p5play.org/v3/p5play.js"></script>
```

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Sprites

The building block of p5play is the **Sprite**:

```
1 function setup() {  
2   createCanvas(400, 400);  
3  
4   // new Sprite(x, y, diameter) -- circle  
5   let ball = new Sprite(200, 100, 50);  
6  
7   // new Sprite(x, y, width, height) -- rectangle  
8   let box = new Sprite(200, 300, 100, 50);  
9 }
```

- Sprites are **automatically drawn**
- They have built-in **physics** – they fall, collide, and bounce

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Gravity and bounciness

```
1 function setup() {  
2   createCanvas(400, 400);  
3   world.gravity.y = 10;  
4  
5   let ball = new Sprite(200, 50, 40);  
6   ball.bounciness = 0.8;  
7   ball.color = "orange";  
8  
9   // "static" sprites don't move  
10  let floor = new Sprite(200, 380, 400, 20, "static");  
11  floor.color = "darkgreen";  
12 }
```

- `world.gravity.y = 10` – things fall down
- `bounciness` from 0 (no bounce) to 1 (perfect bounce)
- `"static"` sprites are fixed in place

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Spawning on click



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What can you make?

- **Sprites/physics** are super useful, even for non-games!
- **Platformer games** – player sprite, platforms, enemies
- **Physics simulations** – balls, chains, ragdolls
- **Interactive art** – physics-driven compositions

Learn more: p5play.org/learn

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Machine Learning

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Why machine learning?

- Your sketch can **see** and **react** to the real world
- Hand tracking, body detection, face recognition – all through the webcam
- Opens up entirely new ways to **interact** with creative work
- The “wow factor” is very high – even with simple code

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What is ml5.js?

- Machine learning made easy for p5.js
- Pre-trained models – no training needed
 - Models are the AI systems
- Combine with your code to add surprising interactions
- ml5js.org

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Including ml5.js

Download into your project folder (preferred):

- [ml5.js](#)

```
1 <script src="ml5.js"></script>
```

Or load from the web:

```
1 <script src="https://unpkg.com/ml5@1/dist/ml5.js"></script>
```

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What models are available?

- **BodyPose** – body pose estimation
- **HandPose** – hand tracking
- **FaceMesh** – face detection
- **ImageClassifier** – recognise objects in images
- **SoundClassifier** – detect sounds
- **(NeuralNetwork** – train your own model)

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Dog breed classifier



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Breaking it down

1. `ml5.imageClassifier("MobileNet")` – load the AI model
2. `classifier.classify(dogImage, gotResult)` – asks the model:
“what breed?”
3. `gotResult(results)` – callback function once the model is done
thinking

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Hand and body tracking

ml5.js can also track your **body in real-time** via the webcam:

- **HandPose** – 21 keypoints per hand (finger tips, knuckles, wrist)
- **BodyPose** – 17 body keypoints (nose, shoulders, elbows, wrists, ...)
- **FaceMesh** – 468 facial landmarks

Same pattern: load model, start detection, draw the keypoints.

These need webcam access – try them locally!

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What can you make?

- **Air drawing** – draw on screen with your finger
- **Gesture controls** – wave to change colours, pinch to zoom
- **Body-controlled games** – move a character with your body
- **Face filters** – overlay graphics on facial landmarks
- **Interactive installations** – art that reacts to the viewer

Learn more: ml5js.org

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Generative Text

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Why generative text?

- Words are a powerful creative medium – not just visuals!
- Generative text creates **unique** output every time
- Grammars let you define the **structure** of language
- Great for poetry, stories, chatbots, and text-based art

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What is RiTa.js?

- A toolkit for **computational writing**
- Generate text from grammars
- Analyse language: rhymes, syllables, parts of speech
- Create Markov chain text (generated text)
- Used by digital poets and artists
- rednoise.org/rita

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Including RiTa.js

Download into your project folder (preferred):

- [rita.min.js](#)

```
1 <script src="rita.min.js"></script>
```

Or load from the web:

```
1 <script src="https://cdn.jsdelivr.net/npm/rita"></script>
```

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Grammars

A grammar defines rules for generating text:

```
1 let grammar = RiTa.grammar({  
2   start: "The $adj $noun $verb",  
3   adj: "[ golden | silent | broken | dancing | ancient ]",  
4   noun: "[ river | shadow | dream | mirror | forest ]",  
5   verb: "[ flows | drifts | whispers | fades | glows ]"  
6 });  
7  
8 let sentence = grammar.expand();  
9 // e.g. "The silent dream whispers"  
10 // e.g. "The golden river flows"
```

- `$adj` gets replaced by a random choice from the `adj` list
- Call `grammar.expand()` to generate a new sentence each time

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Generative poem



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Language analysis

RiTa can also analyse language:

```
1 // Find rhymes
2 let rhymes = RiTa.rhymes("cat");
3 // ["bat", "hat", "sat", "flat", ...]
4
5 // Count syllables
6 let count = RiTa.syllables("creativity");
7 // "kri-ey-tiv-ih-tiy" (5 syllables)
8
9 // Get part of speech
10 let pos = RiTa.pos("The cat sat");
11 // ["dt", "nn", "vbd"] (determiner, noun, verb-past-tense)
```

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What can you make?

- Poetry generators – grammars that write poems
- Interactive fiction – stories that change each reading
- Text visualisations – words as visual elements
- Chatbot personalities – grammar-driven dialogue
- Concrete poetry – where the shape of text matters

Learn more: rednoise.org/rita

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Animated Drawing

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Why animated drawing?

- Static shapes appear instantly – animated ones **tell a story**
- Watching something draw itself feels **alive** and engaging
- Great for presentations, infographics, and storytelling
- Adds polish to any sketch with minimal effort

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What is p5.animS?

- Animates the **drawing process** of p5.js shapes
- Shapes progressively draw themselves over time
- Like watching someone draw them by hand
- Same function names as p5.js, with two extra parameters
- github.com/wixette/p5.animS

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Including p5.anims

Download into your project folder (preferred):

- [p5.anims.min.js](#)

```
1 <script src="p5.anims.min.js"></script>
```

Or load from the web:

```
1 <script src="https://cdn.jsdelivr.net/npm/p5.anims@0.0.7/build/p5.anims.min.js"></script>
```

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Animated circle

```
1 function draw() {  
2   background(220);  
3   // animS.circle(id, duration, x, y, diameter)  
4   animS.circle('c1', 60, 300, 200, 150);  
5 }  
6  
7 function mouseClicked() {  
8   animS.reset(); // replay all animations  
9 }
```

- First parameter: a **unique ID** (any string)
- Second parameter: **duration** in frames
- Rest: same as regular `circle()`

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Multiple animated shapes



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Available shapes

Standard p5.js	p5.animS equivalent
<code>circle(x, y, d)</code>	<code>animS.circle(id, dur, x, y, d)</code>
<code>ellipse(x, y, w, h)</code>	<code>animS.ellipse(id, dur, x, y, w, h)</code>
<code>rect(x, y, w, h)</code>	<code>animS.rect(id, dur, x, y, w, h)</code>
<code>line(x1, y1, x2, y2)</code>	<code>animS.line(id, dur, x1, y1, x2, y2)</code>
<code>triangle(...)</code>	<code>animS.triangle(id, dur, ...)</code>
<code>arc(...)</code>	<code>animS.arc(id, dur, ...)</code>

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Every shape just needs an **id** and **duration** added!

What can you make?

- **Animated infographics** – charts that draw themselves
- **Storytelling sketches** – reveal elements over time
- **Logo animations** – shapes assembling into a design
- **Whiteboard effects** – like someone is drawing live

Learn more: wixette.github.io/p5.animS

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Hand-Drawn Visuals

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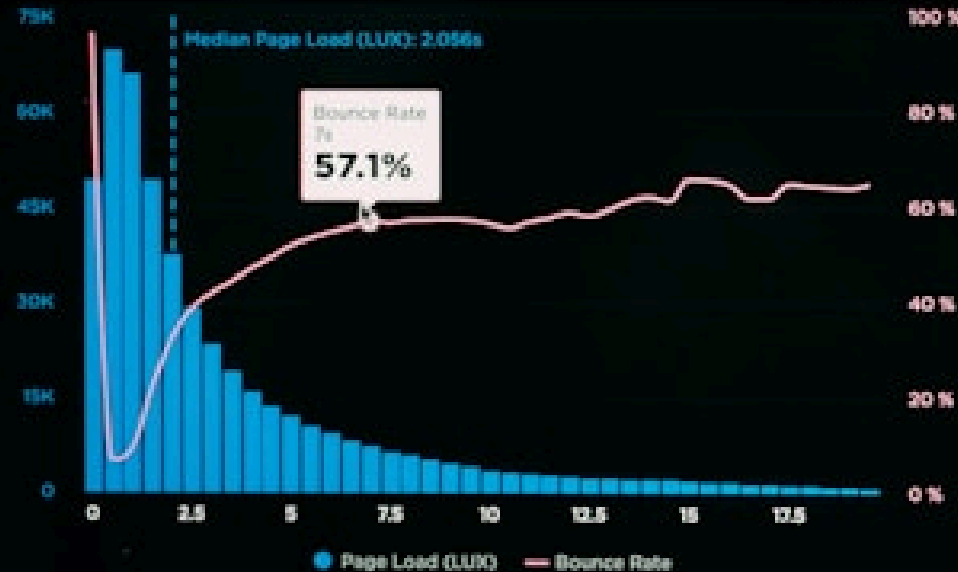


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USERS: LAST 7 DAYS USING MEDIAN ▾

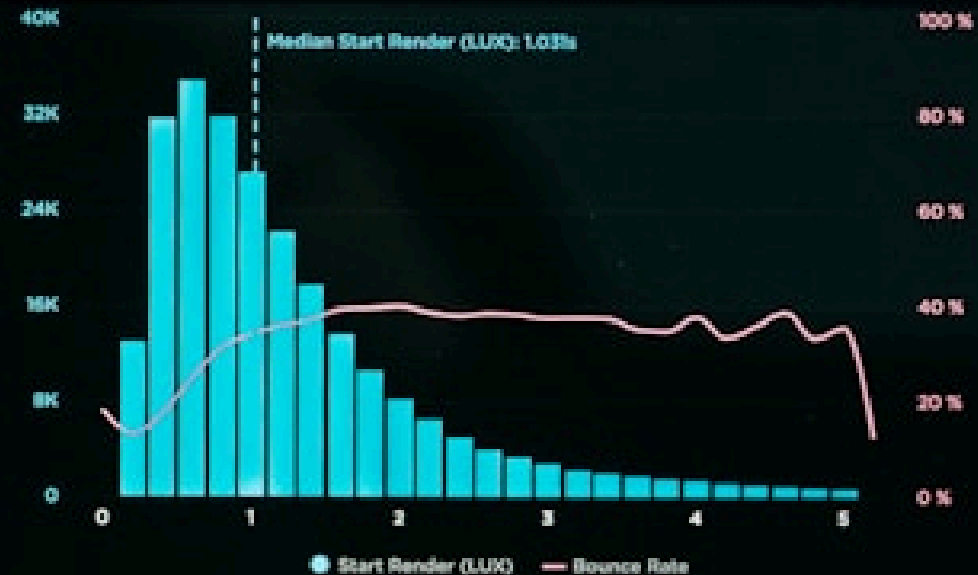
LOAD TIME VS BOUNCE RATE

⊙ OPTIONS



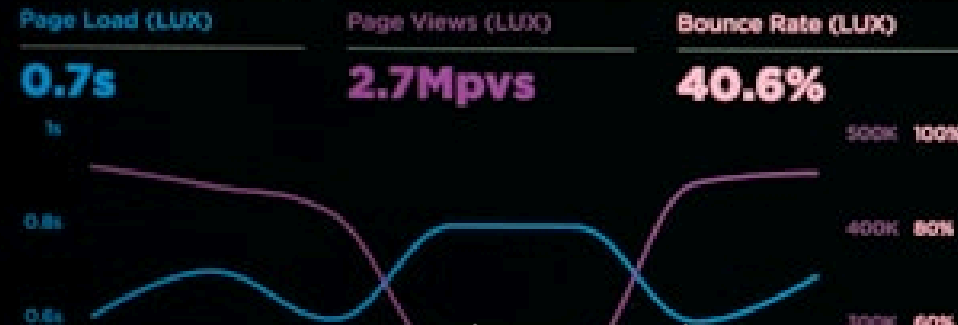
START RENDER VS BOUNCE RATE

⊙ OPTIONS



PAGE VIEWS VS ONLOAD

⊙ OPTIONS



SESSIONS

⊙ OPTIONS



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Why sketchy visuals?

- Hand-drawn visuals feel **personal** and approachable
- They add **character** to data visualisations and graphics
- Transforms any sketch into something that looks hand-made

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What is p5.scribble?

- Replaces standard shapes with **hand-drawn** versions
- Looks like pen on paper
- Just swap `rect()` for `scribbleRect()`
- github.com/generative-light/p5.scribble.js

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Including p5.scribble

Download into your project folder (preferred):

- [p5.scribble.js](#)

```
1 <script src="p5.scribble.js"></script>
```

Or load from the web:

```
1 <script src="https://cdn.jsdelivr.net/gh/generative-light/p5.scribble.js/p5.scribble.js"></s
```

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Basic usage



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Sketchy bar chart



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What can you make?

- **Sketchy data visualisations** – charts with personality
- **Comic-style graphics** – panels, speech bubbles, characters
- **Prototype aesthetics** – wireframe / napkin-sketch look
- **Mixed media** – combine scribble shapes with photos or clean text

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Wrapping up

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Choosing a library

Library	Best for	Complexity
p5play	Games, physics, interactive art	Low
ml5.js	Webcam, body/hand tracking	Low-Medium
RiTa.js	Generative text, poetry	Low
p5.animS	Animated drawing effects	Very low
p5.scribble	Hand-drawn visual style	Very low

You don't have to use any!

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Finding more libraries

- The official list: p5js.org/libraries
- Covers: GUI, particle systems, networking, accessibility, and more
- Each entry links to documentation and examples
- If you can think of it, there's probably a library for it

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Learning and inspiration

- [The Coding Train](#) – video tutorials for p5.js, ml5.js, and more
- [OpenProcessing](#) – community sketches with viewable source code
- [p5js.org/examples](#) – official p5.js examples
- [Generative Design](#) – book with free online examples

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Beyond p5.js

If you want to go further **after this course**:

- [three.js](#) – 3D graphics in the browser
- [Tone.js](#) – advanced audio synthesis
- [matter.js](#) – another physics engine
- [D3.js](#) – powerful data visualisation
- [Cables.gl](#) – visual programming for WebGL

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Summary

- Thank you for joining this course!
- Hopefully the lectures were useful - it was a first time this course setup as well!
- Good luck with your projects!

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